Autism Prediction

Deep Learning

Submitted By:

Anchal 8025320016

Submitted To:

Dr. Sangita Roy

Dr. Seema Bawa

Computer Science and Engineering Department

Thapar Institute of Engineering and Technology

(Deemed to be University), Patiala – 147004

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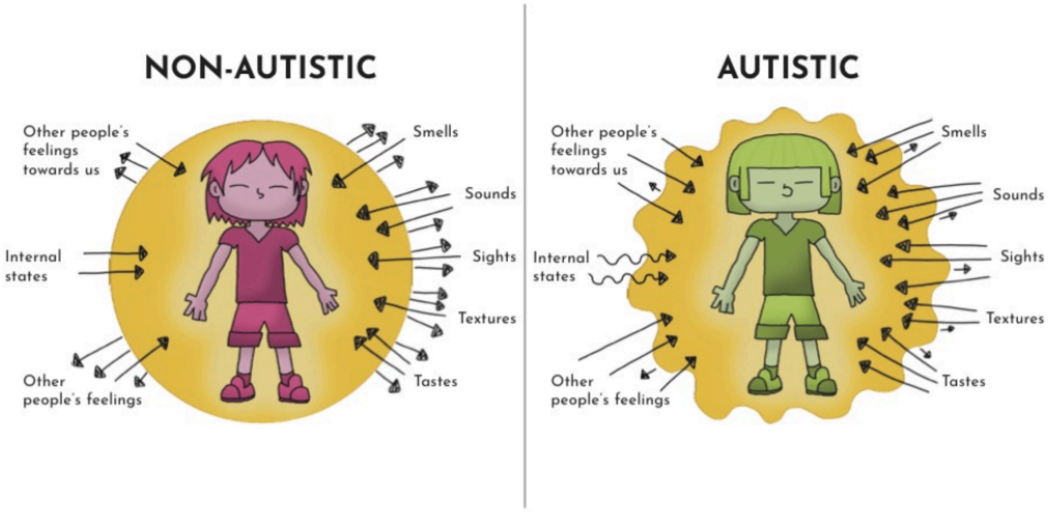
Abstract

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition that affects communication, social interaction, and behavior. Since there is no single medical test to diagnose autism, early detection often depends on behavioral observations and questionnaires such as the AQ-10. This project aims to develop a machine-learning–based screening system that predicts the likelihood of ASD using AQ-10 responses along with demographic and medical indicators. By applying data preprocessing, feature engineering, visualization, model comparison, and interactive deployment via Gradio, the system provides a reliable preliminary tool that aids early autism screening.

1. Introduction

Autism is a neurodevelopmental condition that affects an individual's communication, social interaction, learning patterns, and behavioral responses. Despite its prevalence, there is no single medical test that can definitively diagnose Autism Spectrum Disorder (ASD); diagnosis typically relies on clinical observation and behavioral assessments.



This project explores whether machine learning can assist in early screening by predicting the likelihood of autism based on the AQ-10 questionnaire and key demographic indicators. By analyzing patterns in responses and behavioral scores, the model aims to provide a supportive tool for preliminary ASD detection.

1.1 Autism Adult Dataset

Dataset provided by open ml

https://www.openml.org/search?type=data&status=active&id=428 78

The dataset contains **800 rows and 22 columns**, including AQ-10 scores (A1–A10), demographic factors (age, gender, ethnicity), and the target variable **Autism**(1 = autism, 0 = non-autism).

1.2 System Pipeline Explained

The flowchart illustrates the complete workflow of the autism prediction system. It starts with loading the dataset, followed by data cleaning steps such as handling missing values and converting categorical text into numeric format. After the data is cleaned, exploratory data analysis (EDA) is performed to identify trends, patterns, and class imbalances. Feature engineering then enriches the dataset by creating additional meaningful attributes like the total AQ-10 score and combined risk indicators. The refined data is split into training and validation sets, and class imbalance is addressed through oversampling. Multiple machine learning models are trained and evaluated, with the Linear Regression model achieving the best performance. Finally, the selected model is used to generate predictions, outputting whether autism is likely or unlikely along with a confidence Probability.

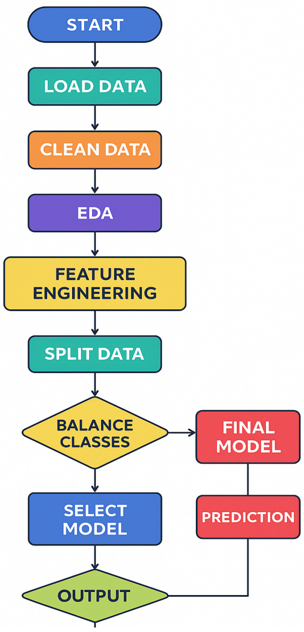


Fig 1 pipeline of the autism prediction system

1.3 Description (800,22)

The provided Autism Adult dataset contains 12 features, each with its own data type and potential missing values. Let's break down the features and their implications.

A1_Score to A10_Score	These likely represent scores from specific autism assessment scales or questionnaires. They are categorical, suggesting different rating levels.
age	Numerical data indicating the age of the individual.
gender	Categorical data representing the gender of the individual.
ethnicity	Categorical data specifying the individual's ethnicity.
jundice	Categorical data indicating whether the individual experienced jaundice.
country_of_res	Categorical data specifying the individual's country of residence.
used_app_before	Categorical data indicating whether the individual used the app before.
result	Numerical data representing the outcome of the assessment.
age_desc	Categorical data describing the age group of the individual.
relation	Categorical data specifying the relationship of the respondent to the individual.
Class/ASD	The target variable, indicating whether the individual was diagnosed with autism (1) or not (0).
austim	Categorical data indicating whether the individual was previously diagnosed with autism.

2. Libraries Used

In this project, several Python libraries were used to handle data processing, visualization, machine learning, model evaluation, UI creation, and file conversion. Each library plays an important role in building the complete autism prediction system:

- **Scipy (scipy.io.arff)** – Used to load and read ARFF files, allowing conversion of raw dataset formats into DataFrames.

- **Pandas** – Helps load, clean, manipulate, and analyze data efficiently using DataFrames.

- **NumPy** – Provides fast numerical operations and supports mathematical functions required for preprocessing and feature engineering.

- **Matplotlib & Seaborn** – Used for data visualization, distribution plots, heatmaps, and other EDA visual insights.

- **Scikit-learn (sklearn)** – Provides tools for preprocessing (LabelEncoder, StandardScaler), splitting the dataset (train_test_split), model development (SVC, LogisticRegression), and evaluation (metrics).

- **Imbalanced-Learn (imblearn.over_sampling)** – Contains RandomOverSampler, used to balance the dataset by oversampling
- the minority class.
- **Warnings** – Helps suppress unnecessary warnings for cleaner output durin

3. Algorithm(s) Used

Support Vector Machine (SVM)

A support vector machine (SVM) is a supervised machine learning algorithm that classifies data by finding an optimal line or hyperplane that maximizes the distance between each class in an N-dimensional space.

Logistic Regression

Logistic Regression is a statistical method used for classification tasks. It predicts the probability of a binary outcome (e.g., yes/no, 0/1). L1 regularization (Lasso) encourages sparsity by setting some coefficients to zero, while L2 regularization (Ridge) penalizes large coefficients, leading to more stable models.

Artificial Neural Network

The Artificial Neural Network (ANN) model was used to classify whether a person has Autism Spectrum Disorder (ASD) or not. The dataset contains around 800 rows and 22 columns, with each row representing one person and each column representing input features such as screening scores, gender, age, and medical details.

The ANN model consists of two hidden layers with 64 and 32 neurons using ReLU activation. Dropout layers were added to reduce overfitting. The output layer uses sigmoid activation for binary classification.

The model was trained using the Adam optimizer and binary cross-entropy loss function for 50 epochs. After training, the model was evaluated using accuracy and ROC-AUC score.

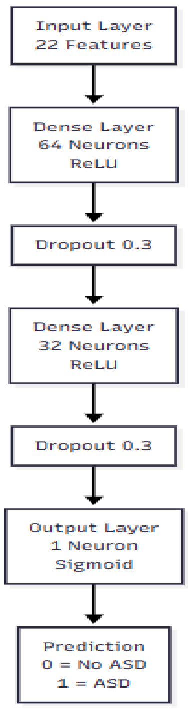


Fig 3.1 ANN

Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) model was also applied to classify ASD and compare its performance with other models. The dataset contains around 800 rows and 22 columns, where each row represents one person.

Since LSTM requires sequence input, the data was reshaped before training. The model includes one LSTM layer with 64 units, a dense layer with 32 neurons, dropout layers to reduce overfitting, and a sigmoid output layer for binary classification.

The model was trained using the Adam optimizer and binary cross-entropy loss function for 50 epochs. Its performance was measured using accuracy and ROC-AUC score.

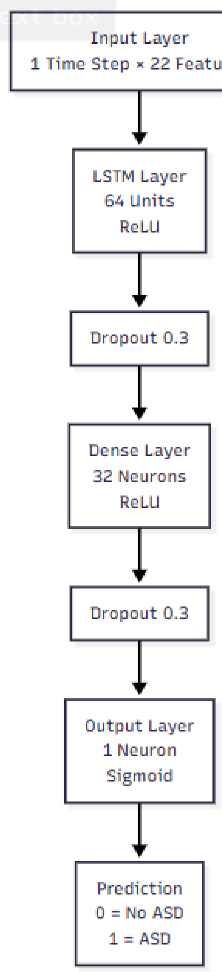


Fig 3.2 LSTM

4. Methodology

4.1 Data Cleaning

- Replace *yes/no* with 1/0.
- Convert “?” and “others” into unified category “Others”.
- Encode categorical features.

4.2 Exploratory Data Analysis (EDA)

- Class imbalance identification
- Distribution plots, countplots, boxplots
- Heatmap to check correlations
- Understanding behavior of AQ-10 scores

4.3 Feature Engineering

- **sum_score:** total of AQ-10 scores
- **ind:** combined medical indicators (autism history + jaundice)
- **log transform of age:** reduce skewness
- **label encoding:** convert text → numbers

4.4 Model Training

After oversampling:

- **Logistic Regression**
- **XGBoost**
- **SVM (RBF kernel) → best model**

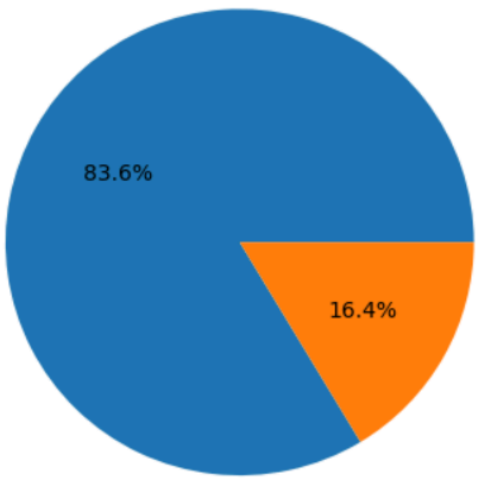
4.5 Deployment

Gradio interface allows real-time ASD prediction using user inputs.

5. Code and Screenshots

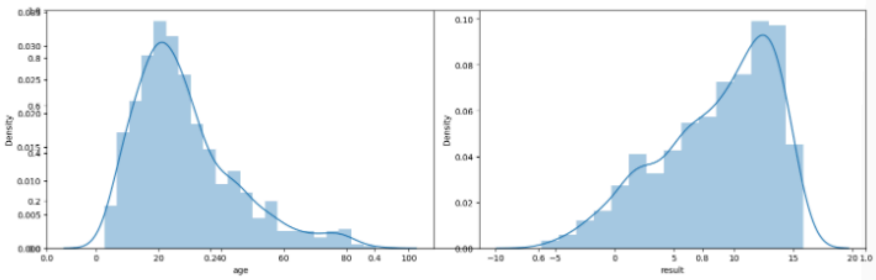
5.1 Exploratory data analysis

```
plt.pie(df['austim'].value_counts().values, autopct='%1.1f%%')
plt.show()
```



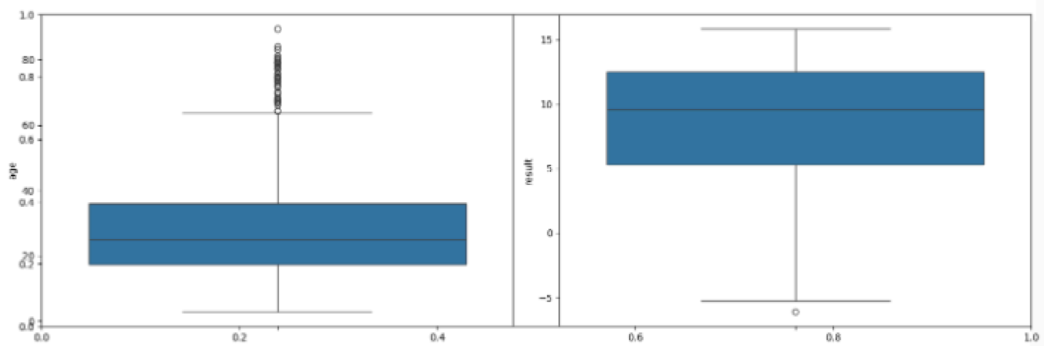
5.2Both of the continuous data are skewed left one is positive and the right one is negatively

```python
import matplotlib.pyplot as plt
import seaborn as sb

floats = ['age', 'result']

plt.subplots(figsize=(15,5))

for i, col in enumerate(floats):
  plt.subplot(1,2,i+1)
  sb.distplot(df[col])
plt.tight_layout()
plt.show()
```

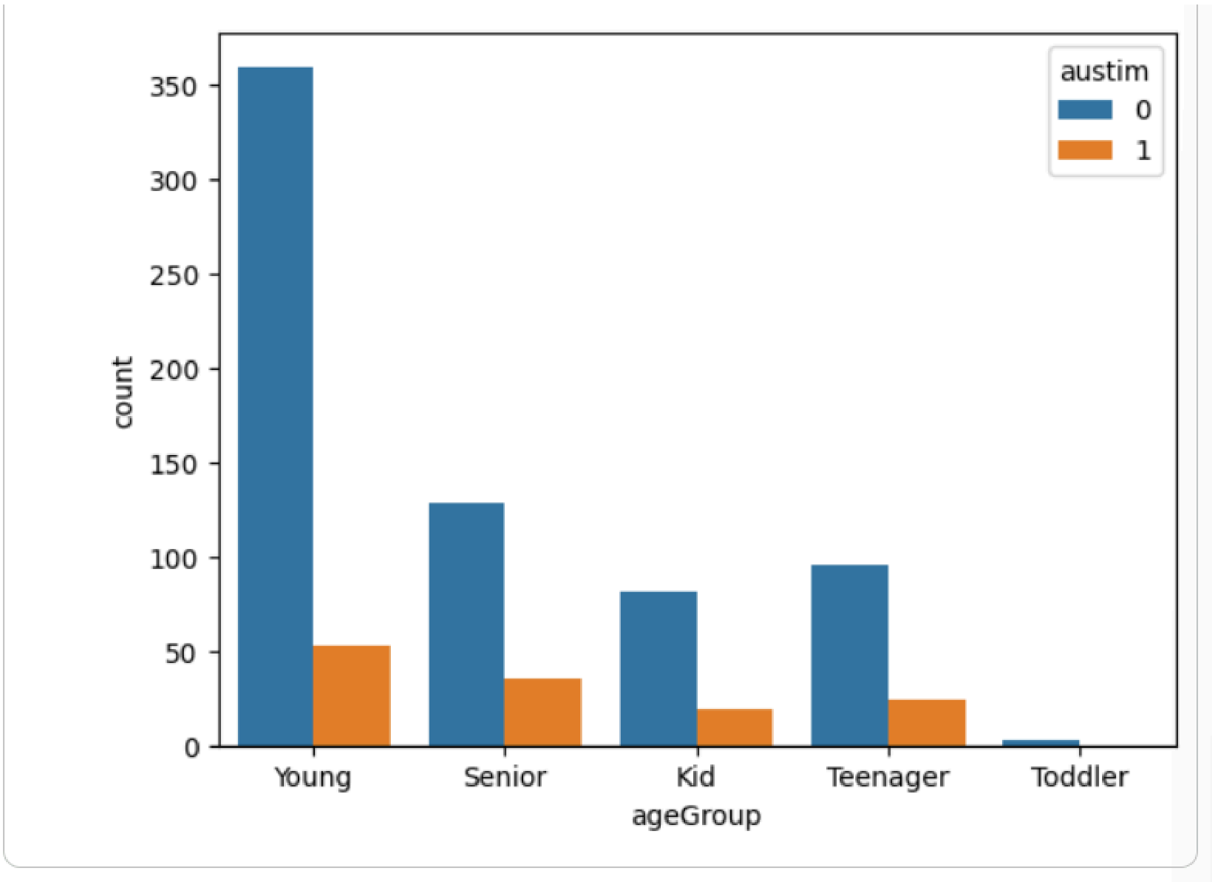


5.3 boxplot

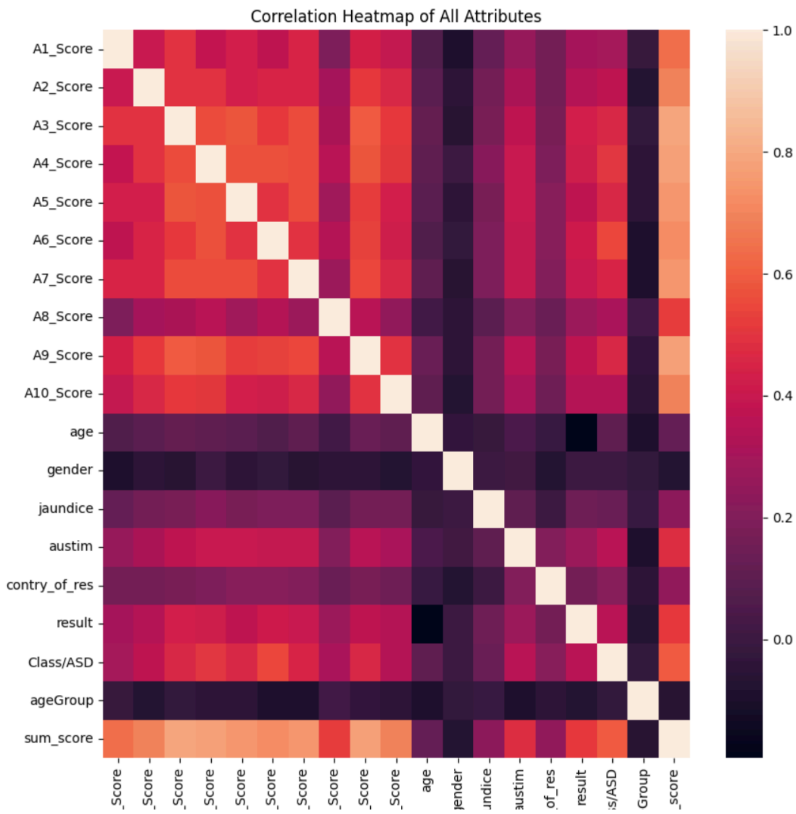
```python
for i, col in enumerate(floats):
  plt.subplot(1,2,i+1)
  sb.boxplot(df[col])
plt.tight_layout()
plt.show()
```



5.4 Feature Engineering



5.6 Heat map



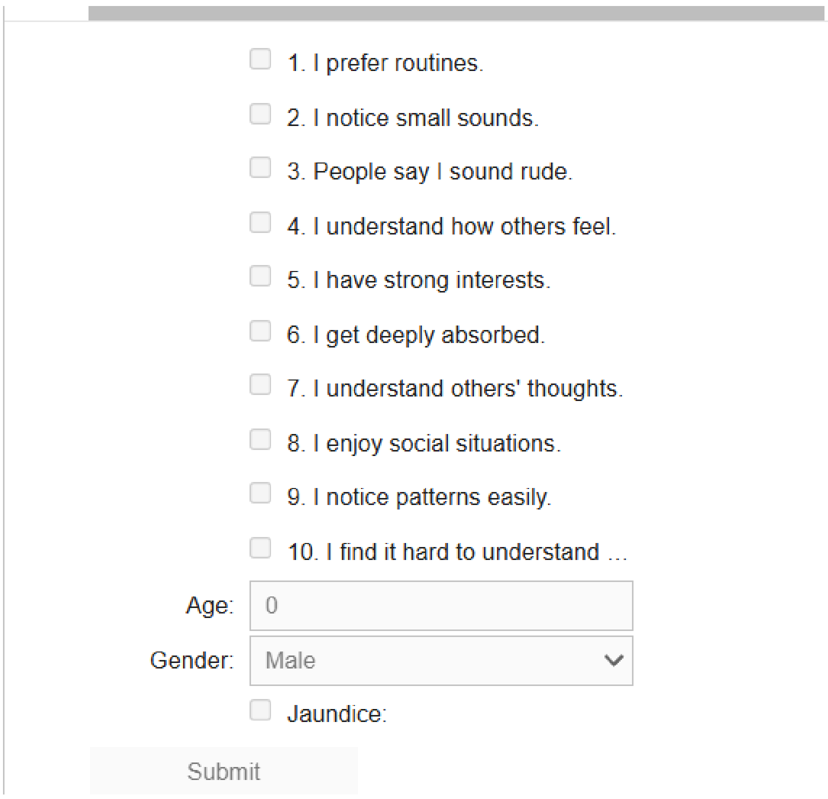
5.7 Now let's train some state-of-the-art machine learning models and compare them

which fit better with our data

Code:

```
--- Updated Model Comparison ---
Logistic Regression:
  Validation Accuracy: 0.7688
  Validation ROC AUC: 0.7815
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step
LSTM:
  Validation Accuracy: 0.7812
  Validation ROC AUC: 0.7563
SVC:
  Validation Accuracy: 0.7688
  Validation ROC AUC: 0.7652
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
ANN:
  Validation Accuracy: 0.7750
  Validation ROC AUC: 0.7689
```

5.8 Interface



6. Results and Discussion

Model Name	Validation Accuracy	Validation ROC AUC
Logistic Regression	0.7688	0.7815
LSTM	0.7750	0.7526
SVC	0.7688	0.7652
ANN	0.7812	0.7726

‘

7. References

[1] Raj, S., & Masood, S. (2020). Analysis and detection of autism spectrum disorder using machine learning techniques. *Procedia Computer Science*, *167*, 994-1004.

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[3] Chowdhury, K., & Iraj, M. A. (2020, November). Predicting autism spectrum disorder using machine learning classifiers. In *2020 International conference on recent trends on electronics, information, communication & technology (RTEICT)* (pp. 324-327). IEEE.